

Overview of the Immune System

A young baby with his mother in India



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The term ***immunity*** *is derived from the Latin word immunitas,*
which referred to the protection from legal prosecution
offered to Roman senators during their tenures in office.

Historically, immunity meant protection from disease and, more specifically, infectious disease.

The cells and molecules responsible for immunity constitute the **immune system, and their collective and coordinated** response to the introduction of foreign substances is called the **immune response.**

- The immune system is remarkably versatile defense system that has evolved to protect animals from invading pathogenic microorganisms and cancer.

- It is able to generate an enormous variety of cells and molecules capable of specifically recognizing and eliminating an apparently limitless variety of foreign invaders.

- These cells and molecules act together in a dynamic network whose complexity rivals that of the nervous system

- Functionally, an immune response can be divided into two related activities—recognition and response.
- Immune recognition is remarkable for its specificity.
- The immune system is able to recognize subtle chemical differences that distinguish one foreign pathogen from another.
- Furthermore, the system is able to discriminate between foreign molecules and the body's own cells and proteins.
- Once a foreign organism has been recognized, the immune system recruits a variety of cells and molecules to mount an appropriate response, called an **effector response,** to eliminate or neutralize the organism.

- In this way the system is able to convert the initial recognition event into a variety of effector responses, each uniquely suited for eliminating a particular type of pathogen.

- Later exposure to the same foreign organism induces a **memory response,** characterized by a more rapid and heightened immune reaction that serves to eliminate the pathogen and prevent disease

Historical Perspective

- The discipline of immunology grew out of the observation that individuals who had recovered from certain infectious diseases were thereafter protected from the disease.

- The Latin term *immunitas*, meaning "exempt," is the source of the English word immunity, meaning the state of protection from infectious disease.

- Perhaps the earliest written reference to the phenomenon of immunity can be traced back to Thucydides, the great historian of the Peloponnesian War.

- In describing a plague in Athens, he wrote in 430 BC that only those who had recovered from the plague could nurse the sick because they would not contract the disease a second time.

- Although early societies recognized the phenomenon of immunity, almost two thousand years passed before the concept was successfully converted into medically effective practice.

- The first recorded attempts to induce immunity deliberately were performed by the Chinese and Turks in the fifteenth century.
- Various reports suggest that the dried crusts derived from smallpox pustules were either inhaled into the nostrils or inserted into small cuts in the skin (a technique called *variolation*).
- In 1718, Lady Mary Wortley Montagu, observed the positive effects of variolation on the native population and had the technique performed on her own children.
- The method was significantly improved by the English physician Edward Jenner, in 1798.
- Intrigued by the fact that milkmaids who had contracted the mild disease cowpox were subsequently immune to smallpox, which is a disfiguring and often fatal disease, Jenner reasoned that introducing fluid from a cowpox pustule into people (i.e., inoculating them) might protect them from smallpox.
- To test this idea, he inoculated an eight-year-old boy with fluid from a cowpox pustule and later intentionally infected the child with smallpox.
- As predicted,the child did not develop smallpox





- Jenner's technique of inoculating with cowpox to protect against smallpox spread quickly throughout Europe.

- Louis Pasteur had succeeded in growing the bacterium thought to cause fowl cholera in culture and then had shown that chickens injected with the cultured bacterium developed cholera.
- After returning from a summer vacation, he injected some chickens with an old culture. The chickens became ill, but, to Pasteur's surprise, they recovered.
- Pasteur then grew a fresh culture of the bacterium with the intention of injecting it into some fresh chickens. But, as the story goes, his supply of chickens was limited, and therefore he used the previously injected chickens.
- Again to his surprise, the chickens were completely protected from the disease.
- Pasteur hypothesized and proved that aging had weakened the virulence of the pathogen and that such an attenuated strain might be administered to protect against the disease.
- He called this attenuated strain a **vaccine** (from the Latin *Vacca*, meaning "cow"), in honor of Jenner's work with cowpox inoculation.

- Pasteur extended these findings to other diseases, demonstrating that it was possible to **attenuate,** or weaken, a pathogen and administer the attenuated strain as a vaccine.
- In a now classic experiment at Pouilly-le-Fort in 1881, Pasteur first vaccinated one group of sheep with heat-attenuated anthrax bacillus (*Bacillus anthracis*); he then challenged the vaccinated sheep and some unvaccinated sheep with a virulent culture of the bacillus.

- All the **vaccinated sheep lived**, and all the unvaccinated animals died.
- These experiments marked the beginnings of the discipline of immunology



- In 1885, Pasteur administered his first vaccine to a human, a young boy who had been bitten repeatedly by a rabid dog (Figure 1-1).

- The boy, Joseph Meister, was inoculated with a series of attenuated rabies virus preparations.

- He lived and later became a custodian at the Pasteur Institute



Wood engraving of Louis Pasteur watching Joseph Meister receive the rabies vaccine

INNATE AND ADAPTIVE IMMUNITY

Defense against microbes is mediated by the early reactions of innate immunity and the later responses of adaptive immunity

Innate **immunity (also called natural or native immunity)** provides the early line of defense against microbes.

It consists of **cellular and biochemical defense** mechanisms that are in place even before infection and are poised to respond rapidly to infections

The mechanisms of **innate immunity** are **specific for structures** that are common to groups of related microbes and **may not distinguish fine differences between microbes.**

The Immune System Includes Innate and Adaptive Components

- Immunity—the state of protection from infectious disease —has both a less specific and more specific component.
- The less specific component, **innate immunity,** provides the first line of defense against infection.
- Most components of innate immunity are present before the onset of infection and constitute a set of disease-resistance mechanisms that are not specific to a particular pathogen but that include cellular and molecular components that recognize classes of molecules peculiar to frequently encountered pathogens.
- Phagocytic cells, such as macrophages and neutrophils, barriers such as skin, and a variety of antimicrobial compounds synthesized by the host all play important roles in innate immunity.
- In contrast to the broad reactivity of the innate immune system, which is uniform in all members of a species, the specific component, **adaptive immunity,** does not come into play until there is an antigenic challenge to the organism.

- Adaptive immunity responds to the challenge with a high degree of specificity as well as the remarkable property of "memory."
- Typically, there is an adaptive immune response against an antigen within five or six days after the initial exposure to that antigen.
- Exposure to the same antigen some time in the future results in a memory response: the immune response to the second challenge occurs more quickly than the first, is stronger, and is often more effective in neutralizing and clearing the pathogen.
- The major agents of adaptive immunity are lymphocytes and the antibodies

Innate Immunity

- Innate immunity can be seen to comprise four types of defensive barriers: anatomic, physiologic, phagocytic, and inflammatory

Type	Mechanism
Anatomic barriers	
Skin	Mechanical barrier retards entry of microbes. Acidic environment (pH 3–5) retards growth of microbes.
Mucous membranes	Normal flora compete with microbes for attachment sites and nutrients. Mucus entraps foreign microorganisms. Cilia propel microorganisms out of body.
Physiologic barriers	
Temperature	Normal body temperature inhibits growth of some pathogens. Fever response inhibits growth of some pathogens.
Low pH	Acidity of stomach contents kills most ingested microorganisms.
Chemical mediators	Lysozyme cleaves bacterial cell wall. Interferon induces antiviral state in uninfected cells. Complement lyses microorganisms or facilitates phagocytosis. Toll-like receptors recognize microbial molecules, signal cell to secrete immunostimulatory cytokines. Collectins disrupt cell wall of pathogen.
Phagocytic/endocytic barriers	Various cells internalize (endocytose) and break down foreign macromolecules. Specialized cells (blood monocytes, neutrophils, tissue macrophages) internalize (phagocytose), kill, and digest whole microorganisms.
Inflammatory barriers	Tissue damage and infection induce leakage of vascular fluid, containing serum proteins with antibacterial activity, and influx of phagocytic cells into the affected area.

The Skin and the Mucosal Surfaces Provide Protective Barriers Against Infection

- Physical and anatomic barriers that tend to prevent the entry of pathogens are an organism's first line of defense against infection.
- The skin and the surface of mucous membranes are included in this category because they are effective barriers to the entry of most microorganisms.
- The skin consists of two distinct layers: a thinner outer layer—the **epidermis**—and a thicker layer—the **dermis.**
- The epidermis contains several layers of tightly packed epithelial cells.
- The outer epidermal layer consists of dead cells and is filled with a waterproofing protein called keratin.

- The dermis, which is composed of connective tissue, contains blood vessels, hair follicles, sebaceous glands, and sweat glands.

- The sebaceous glands are associated with the hair follicles and produce an oily secretion called **sebum.**

- Sebum consists of lactic acid and fatty acids, which maintain the pH of the skin between 3 and 5; this pH inhibits the growth of most microorganisms.

- A few bacteria that metabolize sebum live as commensals on the skin and sometimes cause a severe form of acne.



- Breaks in the skin resulting from scratches, wounds, or abrasion are obvious routes of infection.

- The skin may also be penetrated by biting insects (e.g., mosquitoes, mites, ticks, fleas, and sandflies); if these harbor pathogenic organisms, they can introduce the pathogen into the body as they feed.
- The protozoan that causes malaria, for example, is deposited in humans by mosquitoes when they take a blood meal.

- Similarly, bubonic plague is spread by the bite of fleas, and Lyme disease is spread by the bite of ticks

- The conjunctivae and the alimentary, respiratory, and urogenital tracts are lined by mucous membranes, not by the dry, protective skin that covers the exterior of the body.

- These membranes consist of an outer epithelial layer and an underlying layer of connective tissue.

Saliva, tears, and mucous secretions act to wash away potential invaders and also contain antibacterial or antiviral substances.

- The viscous fluid called mucus, which is secreted by epithelial cells of mucous embranes,entraps foreign microorganisms.
- In the lower respiratory tract, the mucous membrane is covered by **cilia,** hairlike protrusions of the epithelial-cell membranes.

- The synchronous movement of cilia propels mucus-entrapped microorganisms from these tracts.

- In addition, nonpathogenic organisms tend to colonize the epithelial cells of mucosal surfaces. These *normal flora* generally outcompete pathogens for attachment sites on the epithelial cell surface and for necessary nutrients.

- Some organisms have evolved ways of escaping these defense mechanisms and thus are able to invade the body through mucous membranes.

- For example, influenza virus (the agent that causes flu) has a surface molecule that enables it to attach firmly to cells in mucous membranes of the respiratory tract, preventing the virus from being swept out by the ciliated epithelial cells.

- Similarly, the organism that causes gonorrhea has surface projections that allow it to bind to epithelial cells in the mucous membrane of the urogenital tract.